



AQ9.6: Activity Questions 6 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

1 point

Consider a function $f: \mathbb{R}^3 \rightarrow \mathbb{R}$.

Choose the correct statement with respect to ∇f .

☐

∇f has co-domain $\mathbb{R}^2$

☐

∇f has co-domain $\mathbb{R}$

☐

∇f has co-domain $\mathbb{R}^3$

☐

∇f has domain $\mathbb{R}^3$

No, the answer is incorrect.

Score: 0

Accepted Answers:

∇f has co-domain $\mathbb{R}$

∇f has domain $\mathbb{R}^3$

1 point

Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ and suppose its gradient ∇f is continuous. Which of the following statements is true for the directional derivative of f in the direction of $a \in \mathbb{R}^3$? Consider dot product to be the inner product $\langle \cdot, \cdot \rangle$.

☐

$D_a f = \frac{1}{\|a\|} \langle a, \nabla f \rangle$

☐

$D_a f = \langle a, \nabla f \rangle$

☐

$D_a f = \frac{1}{2\|a\|} \langle a, \nabla f \rangle$

☐

$D_a f = \frac{1}{\|a\|^2} \langle a, \nabla f \rangle$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$D_a f = \frac{1}{\|a\|} \langle a, \nabla f \rangle$

Let $f(x, y) = e^x \sin y$. If ∇f at the point $(\ln(2), \pi/4)$ is (k, k) where $k \in \mathbb{R}$, then find the value of k^2 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

Level 2



1 point

Let $p = -\frac{k}{(\|r\|)^3}r$, where $r = (x, y, z) \in \mathbb{R}^3 \setminus \{(0, 0, 0)\}$, $k \in \mathbb{R}$.
 $(x, y, z) \in \mathbb{R}^3 \setminus \{(0, 0, 0)\}$, $k \in \mathbb{R}$.

Amongst the options below, choose those for which the function f satisfies $\nabla f = p \nabla f = p$ for some $k \in \mathbb{R}$.

Consider dot product as the inner product for finding the norm.



$$f(x, y, z) = \frac{1}{\|r\|} f(x, y, z) = \|r\|$$



$$f(x, y, z) = \frac{1}{2\|r\|} f(x, y, z) = 2\|r\|$$



$$f(x, y, z) = \frac{1}{\|r\| + 2} f(x, y, z) = \|r\| + 2$$



$$f(x, y, z) = \|r\| f(x, y, z) = \|r\|$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$f(x, y, z) = \frac{1}{\|r\|} f(x, y, z) = \|r\|$$

$$f(x, y, z) = \frac{1}{2\|r\|} f(x, y, z) = 2\|r\|$$

1 point

Let $f(x, y, z) = 4(x^2 + y^2) - z^2$, $f(x, y, z) = 4(x^2 + y^2) - z^2$.

Choose the correct option(s).



$$\nabla f \nabla f \text{ at } (1, 0, 2) \text{ is } (8, 0, -4)$$



$\nabla f \nabla f$ is continuous everywhere in $\mathbb{R}^3$



$\nabla f \nabla f$ is not continuous everywhere in $\mathbb{R}^3$



$$\nabla f \nabla f \text{ at } (1, 0, 2) \text{ is } (-8, 0, -4)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\nabla f \nabla f \text{ at } (1, 0, 2) \text{ is } (8, 0, -4)$$

$\nabla f \nabla f$ is continuous everywhere in $\mathbb{R}^3$

1 point

Consider a function defined as

$$f(x, y) = \begin{cases} \frac{y^3}{x^2 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases} \quad f(x, y) = \begin{cases} x^2 + y^2 y^3 & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Choose the correct option(s).



$$\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2 + y^2)^2}, \quad \frac{\partial f}{\partial x} = (x^2 + y^2)^2 - 2xy^3, \text{ for } (x, y) \neq (0, 0)$$

☐

$$\frac{\partial f}{\partial x} = \frac{2xy^3}{(x^2 + y^2)^2}, \partial_x \partial f = (x^2 + y^2)^{-2} 2xy^3, \text{ for } (x, y) \neq (0, 0) \quad (x, y) = (0, 0)$$

☐

$$\frac{\partial f}{\partial y} = \frac{3x^2y^2 + y^4}{(x^2 + y^2)^2} \partial_y \partial f = (x^2 + y^2)^{-2} 3x^2y^2 + y^4 \text{ for } (x, y) \neq (0, 0) \quad (x, y) = (0, 0)$$

☐

$$\frac{\partial f}{\partial y} = \frac{3x^2y^2 - y^4}{(x^2 + y^2)^2} \partial_y \partial f = (x^2 + y^2)^{-2} 3x^2y^2 - y^4 \text{ for } (x, y) \neq (0, 0) \quad (x, y) = (0, 0)$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{\partial f}{\partial x} = \frac{-2xy^3}{(x^2 + y^2)^2}, \partial_x \partial f = (x^2 + y^2)^{-2} -2xy^3, \text{ for } (x, y) \neq (0, 0) \quad (x, y) = (0, 0)$$

$$\frac{\partial f}{\partial y} = \frac{3x^2y^2 + y^4}{(x^2 + y^2)^2} \partial_y \partial f = (x^2 + y^2)^{-2} 3x^2y^2 + y^4 \text{ for } (x, y) \neq (0, 0) \quad (x, y) = (0, 0)$$

1 point

Consider the function definition given in Q6 above. Choose the correct option(s).

☐

$\frac{\partial f}{\partial x} \partial_x \partial f$ is continuous at $(0, 0)$ $(0, 0)$

☐

$\frac{\partial f}{\partial x} \partial_x \partial f$ is not continuous at $(0, 0)$ $(0, 0)$

☐

$\frac{\partial f}{\partial y} \partial_y \partial f$ is not continuous at $(0, 0)$ $(0, 0)$

☐

$\nabla f \nabla f$ is discontinuous at the origin $(0, 0)$ $(0, 0)$

☐

Directional derivative of f in the direction of a vector $a \in \mathbb{R}^2$ at $(0, 0)$ can be obtained using the gradient of f

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{\partial f}{\partial x} \partial_x \partial f$ is not continuous at $(0, 0)$ $(0, 0)$

$\frac{\partial f}{\partial y} \partial_y \partial f$ is not continuous at $(0, 0)$ $(0, 0)$

$\nabla f \nabla f$ is discontinuous at the origin $(0, 0)$ $(0, 0)$

1 point

Consider a function defined as

$$f(x, y) = \begin{cases} (x^2 + y^2) \sin\left(\frac{1}{\sqrt{x^2 + y^2}}\right) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases} \quad f(x, y) = \begin{cases} (x^2 + y^2) \sin\left(\frac{1}{\sqrt{x^2 + y^2}}\right) & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$



1) $(x, y) = (0, 0)$ $(x, y) = (0, 0)$.

Choose the correct option(s).

☐

$$\frac{\partial f}{\partial y}(0, 0) = 0 \partial_y \partial f(0, 0) = 0$$

☐

$$\frac{\partial f}{\partial x}(0, 0) = 0 \partial_x \partial f(0, 0) = 0$$

☐

$$\frac{\partial f}{\partial x} = 0 \partial_x \partial f = 0$$

☐

$$\frac{\partial f}{\partial y} = 0 \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x} = 0$$

☐ None of the above

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{\partial f}{\partial y}(0, 0) = 0 \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x}(0, 0) = 0$$

$$\frac{\partial f}{\partial x}(0, 0) = 0 \Rightarrow \frac{\partial}{\partial x} \frac{\partial f}{\partial y}(0, 0) = 0$$

1 point

Consider the function definition given in Q8 above. Choose the correct option(s) with respect to the partial derivative of f . Note $\text{sign}(x) = \pm 1$ depending on the sign of x .

☐

$$\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \text{sign}(x) \cos\left(\frac{1}{|x|}\right) \Rightarrow \frac{\partial}{\partial x} \frac{\partial f}{\partial y}(x, 0) = 2x \sin(|x|) - \text{sign}(x) \cos(|x|) \text{ for } x \neq 0$$

☐

$$\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \text{sign}(x) \cos\left(\frac{1}{|x|}\right) \Rightarrow \frac{\partial}{\partial x} \frac{\partial f}{\partial y}(x, 0) = 2x \sin(|x|) - \text{sign}(x) \cos(|x|)$$

☐

$$\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \cos\left(\frac{1}{|x|}\right) \Rightarrow \frac{\partial}{\partial x} \frac{\partial f}{\partial y}(x, 0) = 2x \sin(|x|) - \cos(|x|) \text{ for } x \neq 0$$

☐

$$\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \text{sign}(y) \cos\left(\frac{1}{|y|}\right) \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x}(0, y) = 2y \sin(|y|) - \text{sign}(y) \cos(|y|) \text{ for } y \neq 0$$

☐

$$\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \text{sign}(y) \cos\left(\frac{1}{|y|}\right) \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x}(0, y) = 2y \sin(|y|) - \text{sign}(y) \cos(|y|)$$

☐

$$\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \cos\left(\frac{1}{|y|}\right) \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x}(0, y) = 2y \sin(|y|) - \cos(|y|) \text{ for } y \neq 0$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$\frac{\partial f}{\partial x}(x, 0) = 2x \sin\left(\frac{1}{|x|}\right) - \text{sign}(x) \cos\left(\frac{1}{|x|}\right) \Rightarrow \frac{\partial}{\partial x} \frac{\partial f}{\partial y}(x, 0) = 2x \sin(|x|) - \text{sign}(x) \cos(|x|) \text{ for } x \neq 0$$

☐

$$\frac{\partial f}{\partial y}(0, y) = 2y \sin\left(\frac{1}{|y|}\right) - \text{sign}(y) \cos\left(\frac{1}{|y|}\right) \Rightarrow \frac{\partial}{\partial y} \frac{\partial f}{\partial x}(0, y) = 2y \sin(|y|) - \text{sign}(y) \cos(|y|) \text{ for } y \neq 0$$

☐

1 point
Consider the function definition given in Q8 above. Choose the correct option.

☐

∇f is continuous everywhere

☐

∇f is discontinuous at the origin $(0, 0)$

☐

$\frac{\partial f}{\partial x}$ is continuous at the origin but $\frac{\partial f}{\partial y}$ is discontinuous at the origin

☐

$\frac{\partial f}{\partial y}$ is continuous at the origin but $\frac{\partial f}{\partial x}$ is discontinuous at the origin

☐

Directional derivative of f in the direction of a vector $a \in \mathbb{R}^2$ at $(0, 0)$ cannot be obtained using the gradient of f

No, the answer is incorrect.

Score: 0

Accepted Answers:

∇f is discontinuous at the origin $(0, 0)$

Directional derivative of f in the direction of a vector $a \in \mathbb{R}^2$ at $(0, 0)$ cannot be obtained using the gradient of f

Your score is: 0/10